Akoizo Original Practice - Form 1 - Answer key

Division B | Food Science | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Lactose is made of which two sugars?

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. Glucose and galactose

B. Glucose and fructose

C. Two glucose units

D. Galactose and fructose

Correct: A

2. (2 points) Casein clumps most at about which pH?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. 9.0

B. 4.6

C. 6.7

D. 7.4

Correct: B

3. (2 points) HTST pasteurization uses:

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. 100 °C for 1 hour

B. 140 °C for 30 minutes

C. 72 °C for 15 seconds

D. 63 °C for 15 seconds

Correct: C

4. (2 points) Which product is safe for someone with lactose intolerance but NOT for someone with milk allergy?

A. Oat milk

B. Water

C. Rice milk

D. Lactose-free cow’s milk

Correct: D

5. (2 points) Forming one triglyceride releases how many water molecules?

Unit: Fats: saturation, esters, and emulsions.
Worked-case data: A team churns 150 mL of cream (mass 148 g) and collects 52 g of butter. If butter is about 80% fat, estimate the cream’s fat percentage.
Separate investigation observations:
Trial 1: Cream at 4 °C: no butter after 15 minutes of shaking.
Trial 2: Cream at 13 °C: butter formed in 8 minutes.
Trial 3: Cream at 25 °C: soft, greasy butter; cloudy buttermilk.

A. Two

B. None

C. Three

D. One

Correct: C

6. (2 points) Which solute lowers freezing point more per mole?

Unit: Freezing point depression and ice cream.
Worked-case data: Estimate the freezing point of meltwater in which 2.0 mol of NaCl dissolves per kilogram of water.
Separate investigation observations:
Trial 1: Milk, 30 mL salt: soft slush, 2 cm tall.
Trial 2: Milk, 90 mL salt: firmer, 4 cm tall.
Trial 3: Cream, 90 mL salt: firm and smooth, 7 cm tall.

A. Sugar (sucrose)

B. They are equal

C. Neither

D. NaCl

Correct: D

7. (2 points) On a label, 25% DV of a nutrient is considered:

A. High

B. Low

C. Moderate only

D. Not meaningful

Correct: A

8. (2 points) A triglyceride contains glycerol and:

Unit: What milk is made of: sugars, fats, and proteins.
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Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. Three amino acids

B. Two glucose units

C. One phosphate and one sugar

D. Three fatty acids

Correct: D

9. (2 points) Why can boiling fresh milk not cause curdling by itself?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. Casein is heat-stable at normal pH

B. Whey proteins cannot denature

C. Boiling raises pH to 10

D. Lactose prevents clumping

Correct: A

10. (2 points) Why does yogurt thicken?

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. Water evaporates

B. Lactic acid lowers pH and casein gels

C. Fat crystallizes

D. Sugar is added

Correct: B

11. (2 points) Each notebook experiment must have:

Unit: Lactose intolerance, allergies, alternatives, and the notebook.
Worked-case data: A student plans to test how vinegar volume affects curd mass. Design one notebook experiment.
Separate investigation observations:
Person 1: Bloating after milk; fine with yogurt and lactose-free milk.
Person 2: Hives within minutes of a tiny amount of cheese.
Test: Person 2 has antibodies to casein.

A. No controlled variables

B. Only one trial

C. Only one independent variable

D. At least three independent variables

Correct: C

12. (4 points) Explain how an emulsifier stabilizes an emulsion.

Unit: Fats: saturation, esters, and emulsions.
Worked-case data: A team churns 150 mL of cream (mass 148 g) and collects 52 g of butter. If butter is about 80% fat, estimate the cream's fat percentage.
Separate investigation observations:
Trial 1: Cream at 4 °C: no butter after 15 minutes of shaking.
Trial 2: Cream at 13 °C: butter formed in 8 minutes.
Trial 3: Cream at 25 °C: soft, greasy butter; cloudy buttermilk.

answer: Its hydrophobic end dissolves in the fat droplet and its hydrophilic end faces the water, forming a coating that keeps droplets from merging. (4 points).

13. (4 points) Explain why freezing point depression is called a colligative property.

answer: It depends only on the number of dissolved particles, not on what the particles are. (4 points).

14. (4 points) Explain why reduced-fat milks are fortified with vitamin A.

Unit: Nutrition and food labels.
Worked-case data: A label lists 300 mg of calcium per serving and a Daily Value of 1,300 mg. A student drinks 3 servings. What percent DV is that, and is one serving "high"?
Separate investigation observations:
Milk A: 8 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk B: 0 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk C: 5 g fat, 24 g total sugars, 12 g added sugars, 25% DV calcium.

answer: Vitamin A is fat-soluble, so it is removed along with the fat and added back to restore the nutrient. (4 points).

15. (4 points) Why are fats insoluble in water?

Unit: What milk is made of: sugars, fats, and proteins.
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Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

answer: Their long hydrocarbon chains are nonpolar, so they cannot form favorable interactions with polar water molecules. (4 points).

16. (4 points) Why should drain time be the same in every trial?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

answer: Curds keep losing water as they drain, so different drain times change the measured mass and make trials incomparable. (4 points).

17. (4 points) Why can many lactose-intolerant people eat yogurt?

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
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answer: Bacteria convert much of the lactose to lactic acid during fermentation, and live cultures may help digest the rest. (4 points).

18. (4 points) Why might soy milk be a better substitute for cow’s milk than almond milk?

answer: Soy milk has protein content close to cow’s milk, while almond milk has very little protein. (4 points).